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The *ice-ice* disease black box: a disease complex triggered by bacterial associations in an *ex situ* milieu of biobanked *Kappaphycus alvarezii*

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Abstract

***Ice-ice* disease (IID) is widely associated with environmental stress and microbial presence in seaweed farms, but its expression under *ex situ* conditions remains poorly understood. This study examined the associated culturable bacterial community in independently maintained *Kappaphycus alvarezii* stock cultures classified as healthy or diseased based on characteristic symptoms. Bacteria were isolated from healthy tissue, diseased tissue, and their corresponding bulk waters, identified using 16S rRNA gene sequencing, and analyzed through phylogenetic reconstruction, species richness estimation, and community similarity metrics. A total of 31 bacterial isolates representing six unique taxa were recovered. Diseased tissue exhibited higher bacterial richness and a distinct tissue-associated bacterial composition compared with healthy tissue. Healthy tissue exclusively harbored *Rossellomorea aquimaris*, whereas diseased tissue contained *Vibrio* and *Photobacterium* species, with representative isolates showing agarolytic and carrageenanolytic activity. In contrast, bulk waters from healthy and diseased cultures contained *Pseudoalteromonas* species. Although the study focuses on microbial associations, the emergence of IID under controlled gene bank conditions highlights the potential interaction of host condition, culture environment, routine handling, and opportunistic bacteria. Together, these factors illustrate the "black box" dynamics of IID in *ex situ* conditions and provide insights for disease monitoring and management in seaweed gene banks.**

Keywords disease complex, eucheumatoids, *ice-ice* disease (IID), Seaweed Culture Laboratory and Gene Bank (SCL&GB)

Introduction

Seaweed gene banks play a particularly important role by preserving valuable germplasm (i.e., the living genetic material of seaweeds typically maintained as branch cuttings), particularly for ecologically and economically significant taxa that are increasingly threatened by climate change, disease outbreaks, and genetic erosion associated with intensive cultivation (Gepts 2006, Araújo et al. 2014, Gonzaga et al. 2025). These gene banks function as specialized systems for the conservation, maintenance, and management of genetic resources, implemented through *ex situ* approaches (Hinaloc and Roleda 2021, Roleda et al. 2024), and function as critical repositories that safeguard biodiversity and support research, breeding, and sustainable resource use. Unlike most seaweed culture laboratories, which commonly maintain specimens for short-term cultivation, e.g., micropropagation and experimentation, seaweed gene banks are specifically dedicated to the long-term storage, conservation, documentation, and management of genetically diverse seaweed resources. In the Philippines, the Seaweed Culture Laboratory and Gene Bank (SCL&GB) at the University of the Philippines, Marine Science Institute (UPMSI) serves both as a culture laboratory for seaweed maintenance and research and as a national seaweed gene bank (Gonzaga et al. 2025). The SCL&GB maintains extensive collections of eucheumatoid seaweeds, a group of carrageenan-producing red seaweeds, comprising farmed cultivars obtained from farming sites across the country and wild specimens collected from natural populations across various Philippine coastal regions. One of the main eucheumatoid species maintained in the SCL&GB is *Kappaphycus alvarezii*, a commercially important carrageenophyte widely cultivated for the production of κ-carrageenan, a sulfated polysaccharide widely used as a gelling and stabilizing agent in food and industrial products (Bui et al. 2019, Rupert et al. 2022). In addition to commonly farmed cultivars, the facility also maintains novel haplotypes representing genetically distinct variants within the species, as well as strains or newly isolated lines derived from either wild populations or from spores released by fertile (cystocarpic) individuals, for subsequent growth performance and biochemical screening (Ganzon-Fortes 2016, Hinaloc and Roleda 2021, Narvarte et al. 2022, Roleda et al. 2024, Gacura et al. 2026). The conservation of such germplasm is essential for sustaining the seaweed industry. It also supports future breeding and

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improvement programs and secures the long-term availability of genetically diverse and resilient stocks (Wade et al. 2020).

Despite their importance, seaweed gene banks are inherently labor-intensive to maintain. Because seaweeds are living organisms, cultures require routine cleaning, frequent water replacement, and continuous supply of sterilized, nutrient-enriched seawater to sustain growth and prevent contamination (Gonzaga et al. 2025). At the SCL&GB, seawater is filtered, sterilized, and supplemented with nutrients using the standard *f/2* enrichment medium (Guillard and Ryther 1962), and cultures are maintained under controlled light, salinity, and temperature conditions to promote stable physiological states and minimize microbial contamination. Nevertheless, disease incidence remains a persistent challenge even under controlled conditions. Among the most frequently observed problems in the facility are *ice-ice* disease (IID) and epiphytism. IID is considered the most destructive disease affecting eucheumatoid seaweed farming worldwide (Largo et al. 1995, Faisan et al. 2021). Although no single standardized case description exists (Ward et al. 2020), IID is typically characterized by localized tissue whitening, followed by softening and eventual thallus disintegration (Trono and Valdestamon 1994, Azizi et al. 2018, Mariño et al. 2019). These symptoms are widely reported in farming areas throughout the Philippines and are associated with substantial biomass loss and significant economic impacts on seaweed farmers (Mendoza et al. 2002, Amilludin 2007, Loureiro et al. 2017, Vairappan et al. 2014). Notably, similar symptoms have also been observed in some cultures maintained within the SCL&GB.

Although IID is widespread and causes substantial impacts, it remains poorly understood, and its precise etiology and infection mechanisms have yet to be fully resolved. The onset of IID has been frequently linked to environmental stressors, including elevated seawater temperatures, abrupt salinity fluctuations, and nutrient imbalances, which are thought to compromise host physiological condition and increase susceptibility to disease (Largo et al. 1995, 2002). In addition to abiotic stress, microbial agents have been strongly implicated in IID development. Several bacterial taxa, particularly within the genera *Vibrio, Alteromonas*, *Pseudoalteromonas*, *Cytophaga*, and *Flavobacterium* have been associated with diseased eucheumatoid thalli (Largo et al. 1995, Aris 2011, Irmawati and Sudirjo 2017, Syafitri et al. 2017). Fungal taxa, including *Aspergillus* and *Phoma* species, have also been reported in association with symptomatic tissues (Solis et al. 2010). However, most of these reported microbial associations have been documented from sea-based farms, where environmental conditions and microbial exposure are inherently variable. This variability complicates the interpretation of causal relationships between specific microorganisms and disease expression. In contrast, the repeated observation of IID-like symptoms in *K. alvarezii* maintained under controlled gene bank conditions, where temperature, salinity, and irradiance levels are regulated, challenges the conventional view that IID is primarily triggered by fluctuating environmental conditions commonly encountered in seaweed farms. Despite long-term maintenance under aseptic and controlled conditions and the selection of visually healthy seedstock, IID symptoms repeatedly developed in some cultures within the SCL&GB. This raises fundamental questions regarding the origin, persistence, and potential role of host-associated bacteria in disease expression under *ex situ* conditions. While most laboratory-based studies on IID have focused on experimental induction or environmental manipulation (Largo et al. 1995, Azizi et al. 2018, Lestari et al. 2026), the present study instead examined IID expression that emerged naturally under gene bank conditions. Concurrently, the operational gene bank environmental conditions and culture maintenance practices were documented. It was hypothesized that healthy and diseased *K. alvarezii* cultures maintained under *ex situ* gene bank conditions harbor distinct culturable bacterial profiles. To test this hypothesis, the present study compared the culturable bacterial communities associated with healthy and diseased *K. alvarezii* stock cultures to identify bacterial taxa uniquely associated with diseased thalli under gene bank conditions. The study specifically focused on culturable bacteria to recover viable bacterial associates that may contribute to disease expression and could be used for future characterization and pathogenicity-related studies. Overall, this study aimed to identify culturable bacterial associations linked to IID in an *ex situ* system and to provide insights that may inform culture management and disease control strategies in seaweed conservation facilities, ultimately supporting the long-term stability and viability of eucheumatoid germplasm in seaweed gene banks.

Materials and methods

Maintenance of gene bank cultures

The SCL&GB follows a general workflow for the maintenance of the eucheumatoid culture collections. Cultures are maintained individually in sterile glass jars containing natural seawater (Fig. 1A). Routine cleaning and water replacement are performed every two weeks (Fig. 1B). For nutrient enrichment, cultures are alternately maintained in sterile seawater during one cleaning cycle and in *f/2*-supplemented seawater during the next. For this study, individual *K. alvarezii* stocks were selected from the culture collection based on visible health status at the time of sampling.

Preparation of culture medium

The seawater used for culture maintenance is sourced from the Bolinao Marine Laboratory (BML), Pangasinan, Philippines, where it is sand-filtered and UV-treated. Upon transfer to the laboratory, the seawater is further filtered through a 1 μm filter bag to remove remaining particulates and subsequently sterilized by autoclaving (121°C, 15 psi, 30 min). The sterilized seawater is then allowed to cool and re-establish equilibrium of its carbonate chemistry before use as the culture medium for routine water replacement. For nutrient enrichment, sterile seawater is supplemented with the standard *f*/2 medium formulation (Guillard and Ryther 1962), a commonly used nutrient medium for marine algal culture.

Sample collection and bacterial isolation

Cultures of *K. alvarezii* of the same strain maintained in individual jars at the SCL&GB were first classified based on the overall condition of the plant present in each culture jar. Each culture jar contained a single *K. alvarezii* plant maintained under the same culture conditions. A culture exhibiting dark, uniform pigmentation, firm texture, and intact branches (thalli) without visible damage or discoloration was classified as healthy (Fig. 2A), whereas a culture showing localized depigmentation or whitening, surface abrasion, softening, and the presence of a characteristic slimy or

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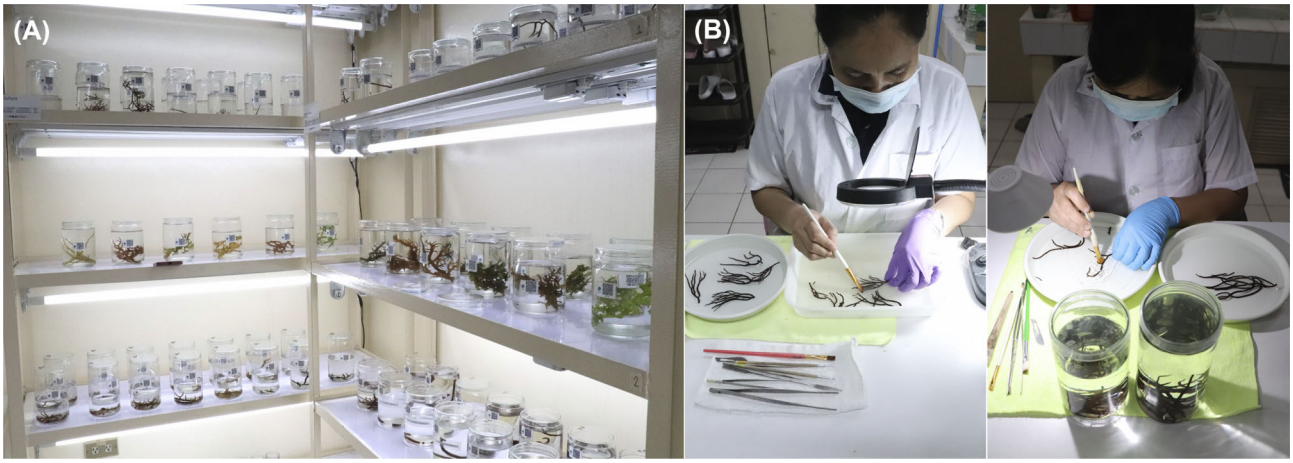


Figure 1 *Ex situ* maintenance of eucheumatoid cultures at the Seaweed Culture Laboratory and Gene Bank (SCL&GB). (A) Eucheumatoid culture collections maintained in individual glass jars under controlled laboratory conditions. (B) Laboratory personnel performing routine manual cleaning of *Kappaphycus alvarezii* thalli using a soft brush as part of the culture maintenance procedure.

mucilaginous film was designated as diseased (Fig. 2B). Based on these characteristics, two culture jars were selected for bacterial isolation: one healthy and one diseased. Because disease occurrence in gene bank is sporadic and unpredictable, sampling focused on a culture that clearly exhibited infection and provided sufficient diseased tissue for bacterial isolation at the time of collection. As such, the study was conducted as a targeted, cross-sectional or snapshot comparison, rather than a replicated experimental design. Following the classification, representative thallus segments were collected from the whole plant in each jar, together with the bulk water. This resulted in four sample types: healthy tissue (HT), water of the healthy plant (HW), diseased tissue (DT), and water of the diseased plant (DW). For the diseased culture, tissue was collected specifically from symptomatic secondary branch of the thallus. For the healthy culture, tissue was also collected from the corresponding secondary thallus branch to minimize variation associated with anatomical sampling location. All samples were processed under aseptic conditions. For bacterial isolation, thallus tissue from healthy and diseased plants was cut into small segments (∼2 mm) using a sterile scalpel and rinsed with sterile seawater to remove loosely attached debris. The clean segments were placed on a sterile tissue paper, covered, and briefly air-dried in a laminar flow hood until excess surface moisture was removed and the thallus segments were no longer dripping. The segments were then carefully inoculated onto Nutrient Agar (NA) plates prepared using seawater. This direct tissue inoculation approach was intended to recover culturable bacteria directly associated with intact thallus tissues. Bulk water samples were serially diluted, and 100 μL aliquots from dilutions (10^{-1} to 10^{-3}) were spread-plated in duplicate onto NA plates. Undiluted bulk water samples were also plated, as the source water was sterile and expected to contain low microbial load. In addition, the sterile seawater used for routine water replacement of cultures was plated as a negative control to monitor potential contamination. All plates were incubated overnight at room temperature (23°C–25°C) and examined for bacterial growth. No additional incubation time was applied beyond the overnight incubation period to prevent fast-growing bacteria from overgrowing the plates and making colony isolation difficult. Colonies exhibiting distinct cultural characteristics were selected, isolated, and purified for Gram-staining, molecular identification, and screening for agarolytic and carrageenanolytic activities.

Screening for agarolytic and carrageenanolytic activities

To identify isolates with agarolytic and/or carrageenanolytic activity, each bacterial isolate was streaked onto agar (Cat. 1802, Condalab) and κ-carrageenan (RICOGEL GU 9801, MRDC) media. The plates were incubated at room temperature (23°C–25°C) for 3 days. After incubation, the plates were flooded with Lugol's iodine solution to visualize any degradation activity, as indicated by the formation of clear zone surrounding the bacterial growth (Azizi et al. 2018).

Molecular identification and taxonomic assignment of bacterial isolates

Pure bacterial cultures were subjected to DNA extraction following the protocol of Chen and Kuo (1993) with minor modifications. The extracted DNA was used as a template for PCR amplification of the 16S rRNA gene region. Each PCR reaction was performed in a 20 μL reaction mixture containing 1 μL of DNA template (10 ng/μL), 1 × Go Flexi PCR buffer, 2 mM $MgCl_2$, 0.2 mM deoxyribonucleotide triphosphates (dNTPs), 0.2 μM of each primer, 1 U GoTaq® G2 Flexi DNA Polymerase (Promega, USA), and nuclease-free water (QIAGEN). Amplification was carried out using a 96-well thermal cycler (TurboCycler Lite, Blue-Ray Biotech), with the universal primers 27F (5'-AGA GTT TGA TCM TGG CTC AG-3') and 1492R (5'-TAC GGY TAC CTT GTT ACG ACT T-3'). Thermal cycling was performed with an initial denaturation of 96°C for 4 min, followed by 30 cycles of denaturation at 94°C for 30 s, annealing at 57°C for 30 s, extension at 72°C for 1 min, and a final extension at 72°C for 10 min (dos Santos et al. 2019). Amplification products were verified by electrophoresis on 1% agarose gel stained with GelRed (Biotium) and visualized using the Blue LED Transilluminator (Labgene Scientific). Successful amplicons were sent to Macrogen Inc. (Seoul, Korea) for bidirectional Sanger sequencing. Raw forward and reverse sequences were pre-processed and assembled using Geneious Prime version 2025.0.2 (Geneious Prime 2025). Consen-

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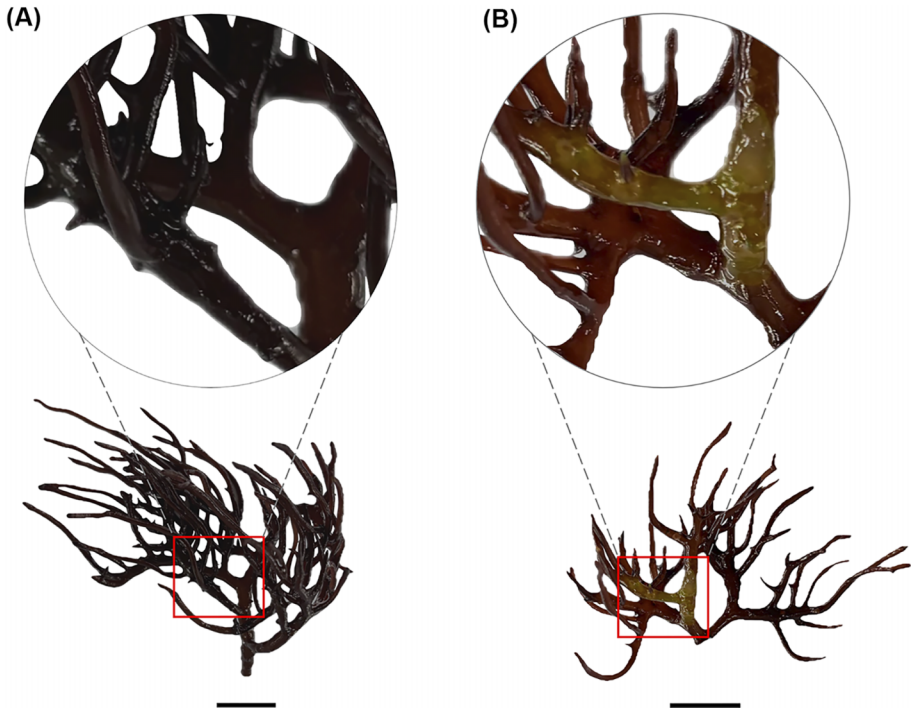


Figure 2 Gross morphology of healthy and diseased *Kappaphycus alvarezii* maintained under gene bank conditions. (A) Healthy specimen showing dark, uniform pigmentation and smooth, intact thalli. (B) Diseased specimen showing localized discoloration, most evident near branch junctions, with weakened and partially disintegrated tissue. Insets highlight magnified regions for detailed visualization of tissue condition. Scale bar = 2 cm.

sus sequences were trimmed and aligned using ClustalW (Thompson et al. 2002) at default setting. Species identification was performed through BLASTn analysis (Altschul et al. 1990) using the NCBI Genbank database as reference. The bacterial species with the highest percentage of similarity was used to assign the taxonomic ID of each bacterial isolate in this study. Taxonomic assignments were used to construct a presence-absence matrix for species richness, community composition, and similarity analyses.

Phylogenetic inference based on 16S rRNA gene sequences

A phylogenetic network of bacterial isolates was constructed, with additional 16S rRNA gene sequences of representative bacterial families included in the final dataset. Sequences were aligned using ClustalW (Thompson et al. 2002) and trimmed to match the length of the shortest sequence. The best-fit nucleotide substitution model was determined using IQ-TREE (Nguyen et al. 2015) based on the Bayesian Information Criterion (BIC) through ModelFinder (Kalyaanamoorthy et al. 2017), and phylogenetic inference was performed using the Maximum Likelihood (ML) method with 1000 bootstrap replicates. The resulting tree was visualized using FigTree (Rambaut 2018).

Species richness and community similarity analyses

The bacterial diversity was evaluated based on species richness (S) per sample type. Richness values were computed from the presence-absence dataset and visualized as bar plots in R version 4.3.1 (R Core Team 2025) using the ggplot2 package to compare richness across sample types. Community composition and similarity were further examined to identify shared and unique taxa. Venn diagrams were generated using the VennDiagram package in R, and pairwise community similarity was calculated using the Jaccard similarity coefficient ($J = a/[a + b + c]$), where a represents shared taxa and b and c denote taxa unique to each pair of samples. The resulting similarity matrix was visualized as a clustered heatmap using the pheatmap package in R.

Results

Taxonomic identification and phylogenetic assignment of bacterial isolates

A total of 31 culturable bacterial isolates were recovered from both healthy and diseased *K. alvarezii* tissues and their bulk waters (Table 1). All isolates were Gram-negative except for a single isolate obtained from the healthy tissue. Notably, two isolates, T1-1 and IS-2, recovered from the diseased tissue and its surrounding water, respectively, exhibited both agarolytic and carrageenanolytic activities (Table 1). Because several isolates shared identical taxonomic IDs, the distinct bacterial taxa detected across the four sample types are summarized in Table 2. In total, six unique culturable bacterial taxa were identified: *Rosselomorea aquimaris*, *Pseudolteromonas* sp., *P.* cf. *piscicida*, *P.* cf. *shioyasakiensis*, *Photobacterium* sp., and *Vibrio* sp. (Table 2). Based on their high sequence similarity to reference strains, some isolates were provisionally labelled with the prefix “cf.” (e.g., *P.* cf.

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Table 1 List of culturable bacterial isolates recovered from healthy and diseased *Kappaphycus alvarezii* tissues and their bulk water with their corresponding taxonomic ID and NCBI Genbank accession numbers.

No.	Strain	Gram reaction	Sample type	Agarolytic activity	Carrageenanolytic activity	Closest match (NCBI)	Percent identity	Assigned ID	Accession no.
1	TH-1	positive	HT	+	–	*Rossellomorea* sp. *Bacillus* sp. *Rossellomorea aquimaris*	100	*Rossellomorea aquimaris*	PX986398
2	H1-1	negative	HW	–	–	*Pseudoalteromonas piscicida* *Pseudoalteromonas* sp.	99.84	*Pseudoalteromonas* cf. *piscicida*	PX986428
3	H1-2	negative	HW	+	–	*Pseudoalteromonas piscicida* *Pseudoalteromonas* sp. *Pseudoalteromonas maricaloris*	96.21	*Pseudoalteromonas* sp.	PX986427
4	H1-3	negative	HW	–	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas lypolytica* *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	100	*Pseudoalteromonas* sp.	PX986426
5	H2-1	negative	HW	+	–	*Pseudoalteromonas piscicida* *Pseudoalteromonas* sp.	100	*Pseudoalteromonas* cf. *piscicida*	PX986425
6	H2-2	negative	HW	–	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas lypolytica* *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	100	*Pseudoalteromonas* sp.	PX986424
7	H2-3	negative	HW	+	–	*Pseudoalteromonas* sp.	99.45	*Pseudoalteromonas* sp.	PX986423
8	H2-4	negative	HW	+	–	*Pseudoalteromonas piscicida* *Pseudoalteromonas* sp.	100	*Pseudoalteromonas* cf. *piscicida*	PX986422
9	H2-5	negative	HW	–	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas lypolytica* *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	98.19	*Pseudoalteromonas* sp.	PX986421
10	H2-6	negative	HW	+	–	*Pseudoalteromonas piscicida* *Pseudoalteromonas* sp.	100	*Pseudoalteromonas* cf. *piscicida*	PX986420
11	T1-1	negative	DT	+	+	*Photobacterium* sp.	99.84	*Photobacterium* sp.	PX986407
12	T1-2	negative	DT	+	–	*Vibrio alginolyticus* *Vibrio parahaemolyticus* *Vibrio* sp.	99.92	*Vibrio* sp.	PX986406

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Table 1 Continued

No.	Strain	Gram reaction	Sample type	Agarolytic activity	Carrageenanolytic activity	Closest match (NCBI)	Percent identity	Assigned ID	Accession no.
13	T1-3	negative	DT	+	–	*Photobacterium* sp.	99.61	*Photobacterium* sp.	PX986405
14	T1-4	negative	DT	+	–	*Photobacterium* sp.	99.92	*Photobacterium* sp.	PX986404
15	T1-5	negative	DT	–	–	*Photobacterium* sp.	99.92	*Photobacterium* sp.	PX986403
16	T1-6	negative	DT	–	–	*Photobacterium* sp.	99.92	*Photobacterium* sp.	PX986402
17	T1-7	negative	DT	–	–	*Photobacterium* sp.	99.92	*Photobacterium* sp.	PX986401
18	T1-8	negative	DT	–	–	*Photobacterium* sp.	99.92	*Photobacterium* sp.	PX986400
19	T1-9	negative	DT	+	–	*Vibrio* sp.	99.53	*Vibrio* sp.	PX986399
20	IS-1	negative	DW	–	–	*Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas* sp. *Ruegeria* sp. *Pseudoalteromonas lypolytica*	100	*Pseudoalteromonas* sp.	PX986409
21	IS-2	negative	DW	+	+	*Vibrio* sp.	99.69	*Vibrio* sp.	PX986408
22	I1-1	negative	DW	–	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas lypolytica* *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	100	*Pseudoalteromonas* sp.	PX986419
23	I1-2A	negative	DW	–	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas lypolytica* *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	100	*Pseudoalteromonas* sp.	PX986418
24	I1-2B	negative	DW	+	–	*Pseudoalteromonas piscicida* *Pseudoalteromonas* sp.	100	*Pseudoalteromonas* cf. *piscicida*	PX986417
25	I1-2C	negative	DW	+	–	*Pseudoalteromonas piscicida* *Pseudoalteromonas* sp.	100	*Pseudoalteromonas* cf. *piscicida*	PX986416
26	I1-3	negative	DW	+	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis*	100	*Pseudoalteromonas* cf. *shioyasakiensis*	PX986415
27	I1-5A	negative	DW	–	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas lypolytica* *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	100	*Pseudoalteromonas* sp.	PX986414

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Table 1 Continued

No.	Strain	Gram reaction	Sample type	Agarolytic activity	Carrageenanolytic activity	Closest match (NCBI)	Percent identity	Assigned ID	Accession no.
28	I1-5B	negative	DW	–	–	*Pseudoalteromonas lypolytica* *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas* sp. *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	97.77	*Pseudoalteromonas* sp.	PX986413
29	I1-6	negative	DW	–	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas lypolytica* *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	100	*Pseudoalteromonas* sp.	PX986412
30	I1-7	negative	DW	+	–	*Pseudoalteromonas piscicida* *Pseudoalteromonas* sp.	100	*Pseudoalteromonas* cf. *piscicida*	PX986411
31	I2-1	negative	DW	–	–	*Pseudoalteromonas* sp. *Pseudoalteromonas shioyasakiensis* *Pseudoalteromonas lypolytica* *Pseudoalteromonas nigrifaciens* *Pseudoalteromonas arabiensis*	100	*Pseudoalteromonas* sp.	PX986410

Sample type: healthy tissue (HT), bulk water of the healthy plant (HW), diseased tissue (DT), and bulk water of the diseased plant (DW); (+) positive for agarolytic or carrageenanolytic activity; (–) negative for agarolytic or carrageenanolytic activity.

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Table 2 Summary of unique culturable bacterial taxa recovered from healthy and diseased *Kappaphycus alvarezii* tissues and bulk water samples under gene bank conditions.

Bacterial taxon	No. of isolate	Gram reaction	Sample type			
			HT	HW	DT	DW
Rossellomorea aquimaris	1	positive	✓			
Pseudoalteromonas cf. *piscicida*	7	negative		✓		✓
Pseudoalteromonas cf. *shioyasakiensis*	1	negative				✓
Pseudoalteromonas sp.	12	negative		✓		✓
Photobacterium sp.	7	negative			✓	
Vibrio sp.	3	negative			✓	✓

A check mark (✓) indicates the presence of the corresponding bacterial taxon for each sample type: healthy tissue (HT), bulk water of the healthy plant (HW), diseased tissue (DT), and bulk water of the diseased plant (DW).

piscicida and *P*. cf. *shioyasakiensis*) to indicate close similarity to known species without confirming exact species identity (Table 1). In the thallus samples, *R. aquimaris* was detected exclusively in healthy tissue (HT), whereas *Photobacterium* sp. and *Vibrio* sp. were identified in diseased tissue (DT). In the bulk water samples, all *Pseudoalteromonas* taxa were confined in the surrounding waters of healthy and diseased cultures (HW and DW), while *Vibrio* sp. was additionally detected in diseased bulk water (DW). No bacteria were isolated from the sterile seawater used for routine culture maintenance, indicating the absence of detectable contamination in the source water. The maximum-likelihood (ML) phylogeny based on the 16S rRNA gene resolved these bacterial isolates into three major clades corresponding to the families Pseudoalteromonadaceae, Vibrionaceae, and Bacillaceae (Fig. 3). The largest group comprised *Pseudolteromonas* species, which formed a well-supported clade. Members of Vibrionaceae, including *Vibrio* and *Photobacterium* species, also clustered into a well-supported clade, while *R. aquimaris* was placed within the Bacillaceae clade.

Culturable bacterial richness and community composition

The species richness of culturable bacteria varied among sample types (Fig. 4A). The healthy tissue (HT) of *K. alvarezii* exhibited the lowest bacterial richness, with only one culturable taxon detected, while its bulk water (HW) contained two taxa. In contrast, the diseased tissue (DT) and its bulk water (DW) harbored two and four taxa, respectively, with DW showing the highest richness among all sample types. When samples were grouped by health status, the combined diseased group (DT + DW) contained five distinct taxa, compared with three taxa in combined healthy group (HT + HW). This observation suggests a higher culturable bacterial richness in diseased samples under the gene bank conditions examined in this study. Furthermore, Venn diagram comparisons showed clear differences in the composition of culturable bacterial communities between healthy and diseased samples (Fig. 4B). No bacterial taxa were shared between HT and DT. In contrast, the bulk waters of healthy and diseased plants (HW and DW) shared two taxa belonging to *Pseudoalteromonas* species, while DW contained two additional unique taxa not detected in HW. The diseased tissue (DT) and its bulk water (DW) contained both unique and shared taxa. *Photobacterium* sp. was detected exclusively in DT, whereas *Pseudoalteromonas* taxa were restricted to DW. In contrast, *Vibrio* sp. was detected in both DT and DW. No culturable taxa were shared between HT and HW. Descriptive similarity comparisons using Jaccard similarity indices (Fig. 4C) further summarized the extent of shared and unique culturable taxa among sample types. The similarity between HT and DT was zero, reflecting the absence of shared culturable taxa. The bulk waters of healthy and diseased plants (HW and DW) showed moderate similarity ($J = 0.50$), corresponding to the two shared *Pseudoalteromonas* taxa. In contrast, similarity between DT and DW was low ($J = 0.20$), reflecting the shared occurrence of *Vibrio* sp. All remaining pairwise comparisons showed zero similarity. Because this study was based on a single healthy and diseased culture, the richness, overlap, and similarity analyses should be interpreted as descriptive exploratory observations rather than inferential community level comparisons.

Discussion

Symptomatology and microbiology of *ice-ice* disease under gene bank conditions

This study demonstrates that *ice-ice* disease (IID) can develop in *K. alvarezii* in *ex situ* gene bank conditions despite stable and controlled environmental settings. Diseased tissue exhibited localized discoloration from dark to light brown near basal regions, often accompanied by the accumulation of mucilaginous film (Fig. 2B). Based on repeated observations during routine culture maintenance, affected areas progressively lose pigmentation, soften, and eventually disintegrate, leading to thallus fragmentation and reduced biomass. These symptoms closely matched classical descriptions of IID reported from sea-based farms (Luxton et al. 1987, Largo et al. 1995, Solis et al. 2010, Rantetondok and Latama 2017, Arasamuthu and Edward 2018, Alibon et al. 2019, Mariño et al. 2019, Riyaz et al. 2020) and laboratory-induced experiments (Largo et al. 1995, Azizi et al. 2018, Lestari et al. 2026). This indicates that the symptoms of IID are similar across different milieu, suggesting that the disease is not limited to field conditions. In-

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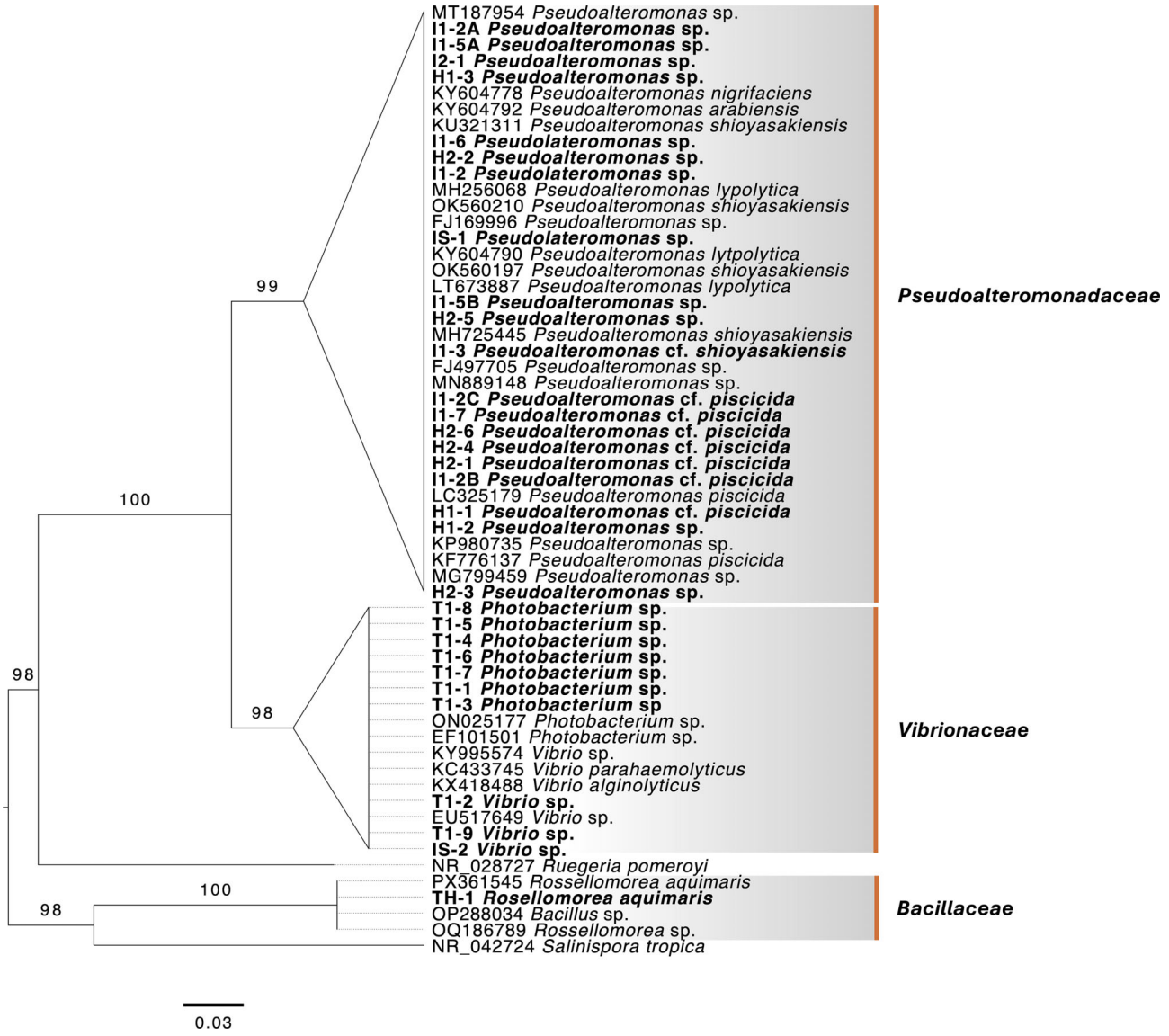


Figure 3 Maximum-likelihood (ML) phylogenetic tree of culturable bacterial isolates associated with healthy and diseased *Kappaphycus alvarezii* maintained under gene bank laboratory conditions, inferred from 16S rRNA gene sequences using IQ-TREE under the TN + F + G4 substitution model. Two marine taxa were included as outgroups to root the tree: *Ruegeria pomeroyi* (Alphaproteobacteria) for the Gammaproteobacteria clade (Pseudoalteromonadaceae and Vibrionaceae), and *Salinispora tropica* (Actinobacteria) for the Firmicutes clade (Bacillaceae). Isolates generated in this study are shown in bold.

stead, IID appears to reflect a general stress response of the host rather than a problem caused only by environmental stochastic events in seaweed farms.

Beyond visible symptoms, the culturable bacterial profiles associated with healthy and diseased thalli revealed distinct bacterial communities under gene bank conditions (Fig. 5). Healthy tissue harbored a Gram-positive bacterium, *R. aquimaris*, whereas diseased tissue was dominated by Gram-negative taxa, including *Vibrio* and *Photobacterium* species. *R. aquimaris* (formerly *Bacillus aquimaris*), a member of the Bacillaceae family (Gupta et al. 2020), has been reported to possess plant growth-promoting properties, probiotic activity, biocontrol potential, and phytoremediation capacity (Yoon et al. 2003, Daryaii et al. 2023, Gupta et al. 2020, Navarro-Torre et al. 2021, Rehan et al. 2023, Paramashivan et al. 2025), and its presence in the healthy tissue suggests that *R. aquimaris* may be beneficial to maintaining the host health. In contrast, the disease-associated taxa *Vibrio* and *Photobacterium*, both belonging to the family Vibrionaceae, have been widely reported as potential pathogens in seaweed manifesting IID symptoms and in other compromised marine host systems (Largo et al. 1995, Labella et al. 2017, Azizi et al. 2018). While *Photobacterium* species are more commonly associated with animal hosts such as corals, sponges, and fish (Labella et al. 2017), their presence in infected *K. alvarezii* tissues in this study expands their documented host range and suggests that they may act as opportunistic bacteria in macroalgal systems. Notably, representative strains of *Vibrio* sp. (IS-2) and *Photobacterium* sp. (T1-1) exhibited both agarolytic and carrageenanolytic activity (Table 1), indicating their ability to degrade agar and carrageenan, which are major components of the algal cell wall. In *K. alvarezii*, carrageenan

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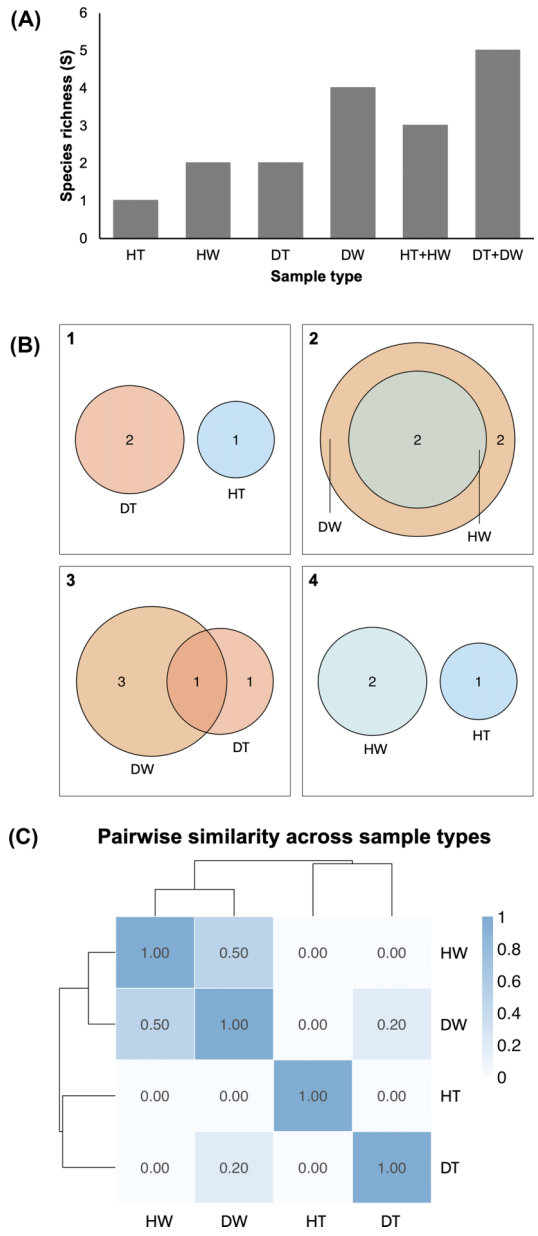


Figure 4 Species richness and compositional relationships of culturable bacterial communities associated with healthy and *ice-ice* diseased *Kappaphycus alvarezii* maintained under gene bank laboratory conditions.(A) Species richness of culturable bacterial taxa isolated from different sample types. Bars represent the number of distinct bacterial species recovered from healthy tissue (HT), bulk water of the healthy plant (HW), diseased tissue (DT), bulk water of the diseased plant (DW), and combined tissue-water samples (HT + HW and DT + DW). (B) Venn diagrams showing the number of unique and shared bacterial taxa between paired sample types: (1) healthy tissue (HT) and diseased tissue (DT); (2) bulk waters of the healthy plant (HW) and diseased plant (DW); (3) diseased tissue (DT) and its bulk water (DW); (4) healthy tissue (HT) and its bulk water (HW). Number within circles represent the count of bacterial taxa unique to or shared between the compared sample types. (C) Jaccard similarity heatmap illustrating pairwise similarities in bacterial community composition across the four sample types (HT, HW, DT, DW). The Jaccard index ranges from 0 to 1, where 0 indicates no shared bacterial taxa and 1 indicates identical community composition. Hierarchical clustering indicates pattern of similarity among sample types.

is the primary structural polysaccharide of the cell wall, making carrageenanolytic activity particularly relevant to tissue degradation. This polysaccharide-degrading activity aligns with how IID symptom progression is commonly observed, which begins with tissue abrasion of the thallus surface, followed by depigmentation or whitening, softening of the cortical and subcortical tissues, and the release of slimy or mucilaginous film. This mucilaginous film is likely associated with the release of cell wall polysaccharides, particularly carrageenan, which may be colonized and utilized by bacteria as a carbon source (Chauhan and Saxena 2016, Zhu and Ning 2016, Azizi et al. 2018, Li et al. 2020, Riyaz et al. 2020, Tahiluddin and Terzi 2021, Rupert et al. 2022). Considering the reported motility of *Vibrio* and *Photobacterium* (Largo et al. 1995, 1999, Azizi et al. 2018), their movement toward and attachment to inadvertently damaged tissues may have initiated infection and IID expression. Moreover, the contrast in bacterial communities between healthy and diseased tissues indicate distinct tissue-associated bacterial composition, with healthy tissue associated with potentially beneficial bacteria and diseased tissue associated with opportunistic taxa. However, because cultures were established without baseline microbiome profiling, it remains unclear whether the bacteria associated with diseased tissue were introduced externally, persisted at low abundance prior to symptom development, or became detectable only after host tissues were compromised.

In addition to tissue-associated bacteria, all *Pseudoalteromonas* species, including *Pseudoalteromonas* sp., *P.* cf. *piscicida*, and *P.* cf. *shioyasakiensis*, were recovered exclusively from the bulk water of both healthy and diseased cultures. Previous studies have shown that *P.* cf. *piscicida* can inhibit competing bacteria, including *Vibrio* and *Photobacterium*, through antimicrobial secretion and direct vesicle-mediated interactions (Richards et al. 2017, Ren et al. 2022). Its detection in the surrounding water of both healthy and infected specimens suggests a potential protective role in the culture environment by limiting the proliferation of potential pathogens. This antagonistic interaction may help explain the absence of *Photobacterium* in the bulk water, although this relationship was not directly tested. However, *Vibrio* sp. was still detected in the bulk water of the symptomatic plant despite the presence of *P.* cf. *piscicida*. A possible explanation is that bacterial cells originating from the diseased tissue may have been present at sufficiently high abundance to persist despite the potential antimicrobial suppression and subsequently disperse into the surrounding medium. Meanwhile, the exclusive recovery of *Pseudoalteromonas* taxa from the bulk waters suggests that they function primarily as environmental generalists within the culture system. Although the seawater used for culture maintenance was sterile at the time of replacement, the occurrence of *Pseudoalteromonas* species in the surrounding water is not fully understood. Because these taxa were not detected in tissue samples, direct tissue association cannot be inferred. As a result, the possibility that *Pseudoalteromonas* taxa were inadvertently introduced during routine culture maintenance cannot be excluded. Members of this genus are ubiquitous in marine environments and are therefore likely to be present in laboratories that routinely handle marine samples (Wrangstadh et al. 1986, Holström and Kjelleberg 1999). Because the preparation area at the SCL&GB is not fully sterile and strict glove use during water replacement and cleaning is not consistently enforced, incidental introduction via non-sterile laboratory surfaces or handling is possible. Airborne transfer through aerosolization is also possible (Michaud et al.

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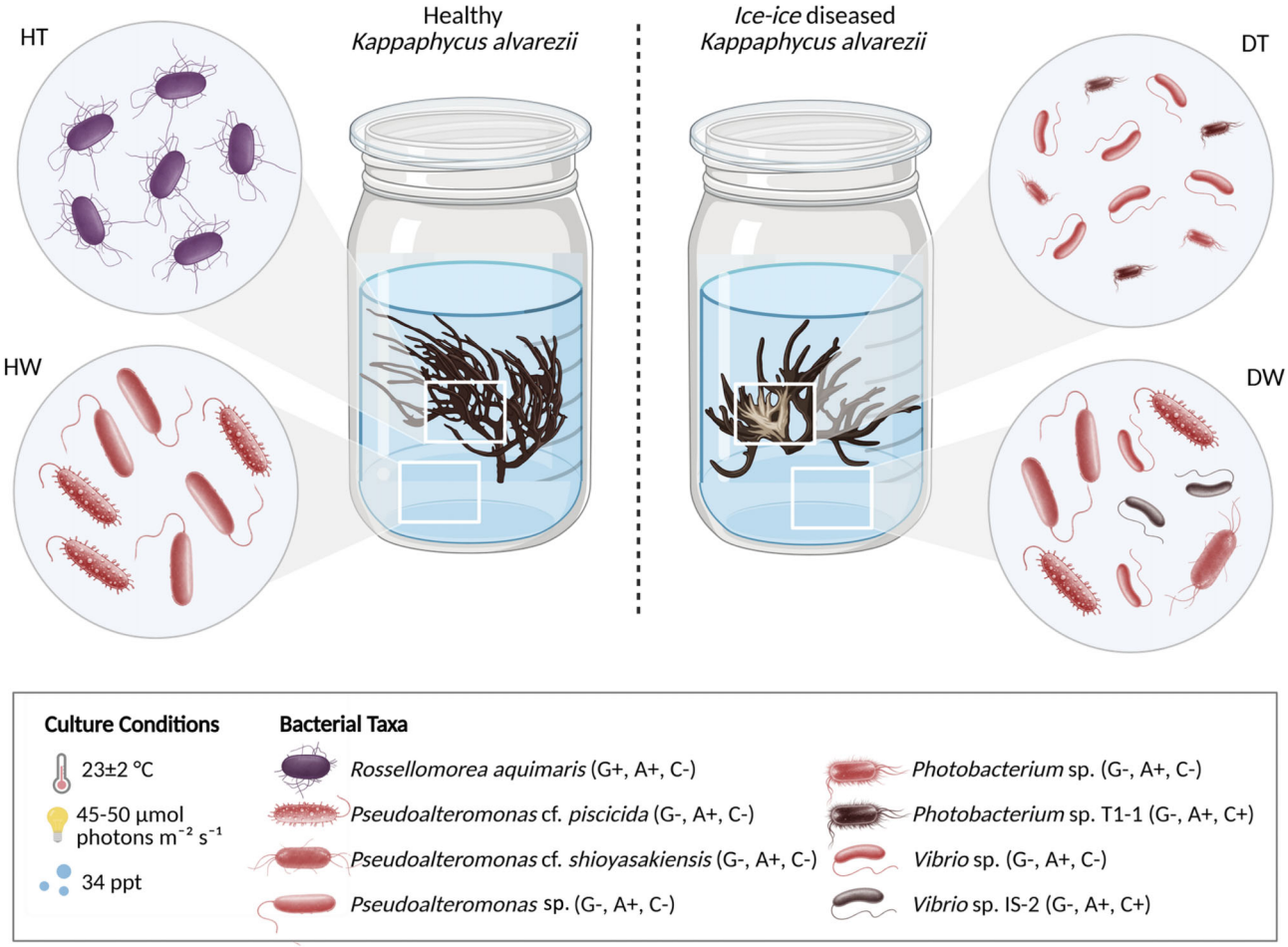


Figure 5 Illustration showing the culturable bacterial communities associated with healthy and *ice-ice* diseased *Kappaphycus alvarezii* stock cultures under gene bank laboratory conditions (23 ± 2°C, 34 ppt salinity, 45–50 µmol photons m^{-2} s^{-1}). Healthy tissue (HT) exclusively harbored the Gram-positive bacterium *R. aquimaris*, whereas diseased tissue (DT) contained *Vibrio* and *Photobacterium* species, including representative strains exhibiting agarolytic and carrageenanolytic activities. Bulk waters from both healthy (HW) and diseased (DW) cultures were dominated by *Pseudoalteromonas* species. Gram-positive (G+), Gram-negative (G–), with agarolytic activity (A+), without agarolytic activity (A–), with carrageenanolytic activity (C+), without carrageenanolytic activity (C–). Illustration created in BioRender. https://BioRender.com/jtaekhv.

2018). Importantly, the absence of *Pseudoalteromonas* species from tissue samples under both healthy and diseased conditions suggests that these taxa do not consistently associate with host tissues. Although *Pseudoalteromonas* has been reported from diseased *Kappaphycus* tissues in previous laboratory culture- and farm-based studies (Syafitri et al. 2017, Azizi et al. 2018), the present findings suggest that its presence in this system may reflect a predominantly environmental or benign association rather than a direct infectious role.

Propensity of gene bank culture conditions to *ice-ice* disease development

In the SCL&GB, cultures are maintained under controlled laboratory conditions, with temperatures of approximately 23 ± 2°C, salinity of 34 ppt, and irradiance levels of 45–50 µmol photons m^{-2} s^{-1} under a 12:12 light-dark photoperiod (Fig. 5). These conditions are intended primarily to support the long-term maintenance and conservation of eucheumatoid germplasm under stable *ex situ* conditions rather than to maximize growth or biomass production. Based on previous experimental studies, temperature and salinity at the SCL&GB fall within the reported optimal range for *Kappaphycus* growth, whereas irradiance levels are lower than those considered optimal (Trono and Ohno 1989, Largo et al. 1995). Laboratory induction experiments have shown that tissue whitening can be triggered by elevated temperatures (33–35 °C), low irradiance (<50 µmol photons m^{-2} s^{-1}), and reduced salinity (<20 ppt), while optimal laboratory conditions (25°C, 125 µmol photons m^{-2} s^{-1}, and 34 ppt salinity) do not induce disease (Trono and Ohno 1989, Largo et al. 1995). Within this context, the SCL&GB culture environment can be considered generally optimal but potentially suboptimal in terms of light availability. While the irradiance levels may not be sufficient to uniformly induce symptoms of IID, they may impose sublethal physiological stress (Trono and Ohno 1989, Largo et al. 1995). This interpretation is supported by the observation that only a subset of cultures developed tissue whitening under otherwise uniform culture conditions, indicating variable stress responses among individual plants. Such variability may be due to differences in thallus age, physiological condition, microbial load, or localized surface damage (Egan et al. 2013). In the context of gene bank management, this cre-

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ates a "black box" scenario: cultures that initially appear healthy are maintained under controlled conditions, yet overtime some remain healthy while others develop symptoms of IID. Because baseline physiological and microbiome data were not collected at the time of culture establishment, it cannot be determined how host condition changed before symptoms appeared. Thus, the observed symptoms of IID likely reflect the combined influence of culture conditions, host stress, and opportunistic microbes rather than a single causative factor.

Potential contribution of routine culture handling to *ice-ice* disease susceptibility

In the SCL&GB, cultures are periodically cleaned using soft brushes to remove surface grime, diseased tissue, and epiphytes (Fig. 1B). While this procedure is essential for maintaining culture quality, repeated mechanical cleaning may inadvertently cause microabrasions or disruption of the thallus surface. Such surface damage provides entry points for opportunistic bacteria to attach and penetrate the cortex and medulla cells (Egan et al. 2014). Once access is gained, carrageenanolytic bacteria such as the *Vibrio* and *Photobacterium* may further degrade host tissues, thereby accelerating tissue softening and disintegration (Azizi et al. 2018, Riyaz et al. 2020, Tahiluddin and Terzi 2021). Although this study did not experimentally test the effects of cleaning-induced stress, the co-occurrence of mechanical handling, tissue degradation, and infection-associated bacteria suggests that routine maintenance practices may act as an additional stressor that facilitates microbial invasion under otherwise controlled conditions. This highlights the need to consider physical disturbance and tissue damage, alongside microbial and environmental factors in understanding IID development in *ex situ* culture systems. Although the effects of cleaning intensity and technique were not directly evaluated in this study, variation in routine handling practices among laboratory personnel may introduce differences in mechanical stress experienced by cultures. Establishing standardized culture guidelines, such as clearly defined criteria for when cleaning is required, recommended brushing techniques and approximate limits on brushing duration or applied pressure, could help reduce unintended tissue stress and improve consistency in culture maintenance.

Conclusion, limitation, and implication for disease management in gene bank cultures

The findings from this study suggest that IID may represent a disease complex shaped by interacting microbial, environmental, and physical factors, rather than a condition driven by a single etiological agent. The demonstration of polysaccharide-degrading activity provides an important functional link between disease-associated taxa and the hallmark feature of IID; however, it does not establish whether these bacteria initiate tissue damage or proliferate in response to pre-existing host stress. To examine this further, future studies should include baseline microbiome profiling, time-series sampling, and experimental manipulation of environmental conditions to better understand whether changes in these factors cause IID. Nevertheless, the results of the present study should be interpreted in the context of the limited number of cultures available for comparison at the time of sampling. Because this study relied on culture-based techniques, the results are limited to culturable bacterial taxa. In addition, the use of a single general-purpose medium may have limited the recovery of more fastidious or specialized marine-associated bacteria. Future work should include pathogenicity assays to directly test the roles of infection-associated bacteria in disease development (Azizi et al. 2018). In addition, culture-independent approaches such as metabarcoding (Campbell et al. 2015, Zozaya-Valdes et al. 2015, Dobretov et al. 2021) or shotgun metagenomic sequencing (Fan et al. 2013, Zozaya-Valdes et al. 2017) are recommended to capture the full microbial diversity associated with IID, including microorganisms that cannot be detected using culture-dependent methods. Also, this study focused only on *K. alvarezii*, so it should be interpreted in the context of this species, as host-microbe interactions may vary among seaweed taxa (Saha et al. 2020, Wood et al. 2022). Future studies including other commercially important eucheumatoid species, such as *K. striatus* and *Eucheuma denticulatum*, would help determine whether similar bacterial associations occur across related hosts. Overall, the occurrence of IID under gene bank conditions highlights the need to integrate several considerations into routine culture management. First, light regimes should be carefully evaluated and adjusted toward optimal irradiance levels to reduce possible physiological stress. Second, routine cleaning practices, while essential for controlling epiphytes, should be optimized to reduce mechanical damage to thallus surfaces through gentler or less frequent handling. The exclusive association of *R. aquimaris* with healthy tissue may suggest that the loss of health-associated bacteria could serve as an indicator of declining thallus condition. Conversely, the appearance of *Vibrio* and *Photobacterium* species, particularly those exhibiting polysaccharide-degrading activity, may signal active disease progression. Finally, maintaining stable water quality and minimizing cross-contamination between culture vessels are essential to limit microbial spread following tissue breakdown. Together, these strategies emphasize that effective gene bank management should address not only environmental parameters but also physical handling and microbial dynamics to secure the long-term health and stability of conserved eucheumatoid germplasm.

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Conflicts of interest

The authors declare that they have no potential conflicts of interest.

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Data availability

All relevant data for the bacterial isolates are provided in Table 1. The nucleotide sequences generated in this study have been deposited in the National Center for Biotechnology Information (NCBI) GenBank, and their corresponding accession numbers are listed in the same table. The sequences can also be accessed directly through NCBI under the accession range: PX986398–PX986428 https://www.ncbi.nlm.nih.gov/nuccore?term=PX986398:PX986428[Accession].

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